industries, but not as widely known as you would expect for such a cornerstone.

In section 4.1, we introduce the problem that will be our North star in this chapter: we need to keep track of large entities, with the smallest memory print possible.

In section 4.2 we continue our narration by discussing a few increasingly complex solutions, showing their strengths and weaknesses: the latter, in particular, ought to be considered chances of improvements, fertile ground for algorithms designers.

As part of this discussion, we introduce the *dictionary*, an abstract data type that we discuss in-depth in section 4.3, while section 4.4 switches to concrete data structures that implements dictionaries: hash tables, binary search trees, and bloom filters.

You have probably guessed that we are particularly interested in the latter, being the topic of this chapter: we describe the principles governing how a bloom filter work in section 4.5, and then delve into each of its methods in section 4.6, showing the pseudo-code for the crucial parts.

Section 4.7 closes the first part of the chapter with a discussion on some typical use cases for Bloom filters: from coolest distributed databases and file systems to rooters, this technology is pervasive.

In order to reach this goal, we keep a practical approach with an emphasis on enabling readers to recognize opportunities